

COMPREHENSIVE SALT ANALYSIS GUIDE

Systematic Qualitative Analysis of Acidic (Anions) & Basic (Cations) Radicals

PART 1: TESTING FOR ACIDIC RADICALS (ANIONS)

Acidic radicals are classified based on their reaction with Dilute H_2SO_4 , Concentrated H_2SO_4 , or specific reagent precipitations.

Radical	Preliminary Reagent	Observations & Gas Evolved	Confirmatory Tests
Sulfite (SO_3^{2-})	Salt + dil. H_2SO_4	- Colorless gas evolves with a characteristic pungent smell of burning sulfur (SO_2). - Turns moist blue litmus paper red.	$KMnO_4$ Test: Pass gas through acidified $KMnO_4$ solution → Pink color decolorizes. $K_2Cr_2O_7$ Test: Pass gas through acidified $K_2Cr_2O_7$ solution → Orange solution turns green.
Sulfide (S^{2-})	Salt + dil. H_2SO_4	- Colorless gas evolves with a foul rotten egg smell (H_2S). - Turns moist blue litmus paper red.	- Lead Acetate Test: Pass gas over paper dipped in $Pb(CH_3COO)_2$ solution → Turns silvery black / shiny black (PbS ppt). - Sodium Nitroprusside Test: Salt solution + fresh sodium nitroprusside → Deep purple/violet color.
Nitrite (NO_2^-)	Salt + dil. H_2SO_4	Reddish-brown gas (NO_2) with a pungent smell evolves on mild reaction.	- Ferrous Sulfate Test: Salt Solution + CH_3COOH + fresh $FeSO_4$ solution → Dark brown / black coloration. - Starch-Iodide Test: Salt Solution + dil. H_2SO_4 + KI solution + starch solution → Deep blue coloration.
Chloride (Cl^-)	Salt + conc. H_2SO_4	Colorless gas with a pungent, suffocating smell (HCl) forms dense white fumes with glass rod dipped in NH_4OH.	NH_4OH Glass Rod Test: Dense white fumes of NH_4Cl at the mouth of test tube. - Silver Nitrate Test: Salt Solution + $AgNO_3$ solution → Curdy white ppt ($AgCl$), soluble in excess NH_4OH. - Chromyl Chloride Test: Salt + $K_2Cr_2O_7$ (powder) + conc. H_2SO_4 → Heat → Orangish-yellow gas (CrO_2Cl_2). Pass gas into NH_4OH → Yellow solution. Add Lead Acetate + CH_3COOH → Yellow ppt ($PbCrO_4$).
Bromide (Br^-)	Salt + conc. H_2SO_4	Reddish-brown gas (Br_2) evolves with a pungent, irritating odor. Intensely colored upon	Silver Nitrate Test: Salt Solution + $AgNO_3$ solution → Pale yellow ppt ($AgBr$), sparingly/partially soluble in

		heating.	NH_4OH. 2. Organic Layer Test: Salt Solution + CCl_4 (or CHCl_3) + conc. HNO_3 (or chlorine water) → Shake vigorously → Organic layer turns orange-brown due to dissolved Br_2.
Nitrate (NO_3^-)	Salt + conc. H_2SO_4 + Paper Ball	Light reddish-brown fumes evolve upon heating; fumes become thick & dark on adding a paper pellet (due to NO_2 evolution).	1. Brown Ring Test: To Salt Solution, add equal volume of freshly prepared FeSO_4 solution. Mix well. Carefully add conc. H_2SO_4 along the inner wall of test tube without disturbing → A distinct dark brown ring forms at the junction of the two liquid layers $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]\text{SO}_4$.
Sulfate (SO_4^{2-})	Independent Test (Water Extract)	No specific gas evolved with dil. or conc. acids.	1. Barium Chloride Test: Salt Solution + BaCl_2 solution → White ppt (BaSO_4), insoluble in dil. HCl or dil. HNO_3. 2. Lead Acetate Test: Salt Solution + $\text{Pb}(\text{CH}_3\text{COO})_2$ solution → White ppt (PbSO_4), soluble in hot ammonium acetate.
Phosphate (PO_4^{3-})	Independent Test (Water Extract)	No gas evolved with dil./conc. H_2SO_4.	1. Ammonium Molybdate Test: Salt Solution + conc. HNO_3 → Boil → Add excess Ammonium Molybdate reagent → Canary yellow precipitate forms (Ammonium phosphomolybdate).

PART 2: SYSTEMATIC ANALYSIS OF BASIC RADICALS (CATIONS)

Basic radicals are classified into Groups 0 through VI based on their qualitative separation scheme using specific Group Reagents.

Group	Cations Included	Group Reagent	Observation / Precipitate Formed
Group 0	NH_4^+ (Ammonium)	NaOH solution + Heat	Pungent smell of Ammonia gas (NH_3). Gives dense white fumes with HCl rod & brown ppt with Nessler's Reagent.
Group I	Pb^{2+} (Lead)	Dilute HCl	White precipitate of Lead Chloride (PbCl_2). Soluble in hot water.

Group II	Cu²⁺ (Copper), Pb²⁺	Dilute HCl + H ₂ S gas	Black precipitate (CuS / PbS). Cu ²⁺ solution is blue; gives chocolate brown ppt with K ₄ [Fe(CN) ₆].
Group III	Al³⁺ (Aluminum), Fe³⁺ (Iron)	NH ₄ Cl (s) + NH ₄ OH (aq) in excess	Al ³⁺ : Gelatinous white ppt [Al(OH) ₃]. Gelatinous white ppt gives Blue Lake test. Fe ³⁺ : Reddish-brown ppt [Fe(OH) ₃].
Group IV	Zn²⁺ (Zinc), Ni²⁺ (Nickel), Co²⁺ (Cobalt), Mn²⁺ (Manganese)	NH ₄ Cl + NH ₄ OH + H ₂ S gas	Zn ²⁺ : Dull white/greyish ppt (ZnS). Ni ²⁺ : Black ppt (NiS) → Crimson red ppt with DMG. Co ²⁺ : Black ppt (CoS). Mn ²⁺ : Buff/flesh-colored ppt (MnS).
Group V	Ba²⁺ (Barium), Sr²⁺ (Strontium), Ca²⁺ (Calcium)	(NH ₄) ₂ CO ₃ + NH ₄ OH + NH ₄ Cl	White precipitate of Carbonates [BaCO ₃ , SrCO ₃ , CaCO ₃]. Ba ²⁺ gives yellow ppt with K ₂ CrO ₄ . Ca ²⁺ gives white ppt with Ammonium Oxalate.
Group VI	Mg²⁺ (Magnesium)	NH ₄ OH + Disodium hydrogen phosphate [(NH ₄) ₂ HPO ₄]	White crystalline precipitate of Magnesium Ammonium Phosphate [Mg(NH ₄)PO ₄].

PART 3: CONFIRMATORY TESTS FOR COMMON BASIC RADICALS

Cation	Confirmatory Test Procedure	Key Observation & Inference
Ammonium (NH₄⁺)	1. Salt + NaOH solution → Heat gas evolved. 2. Pass gas / add solution to Nessler's Reagent K ₂ [HgI ₄].	1. Ammonia smell + white dense fumes with rod dipped in conc. HCl. 2. Brown precipitate or reddish-brown coloration.
Lead (Pb²⁺)	1. Salt solution + Potassium Iodide (KI) solution. 2. Heat the KI precipitate with water and cool.	1. Bright yellow precipitate of Lead Iodide (PbI ₂). 2. Golden spangles / glittering yellow crystals form upon cooling (Golden Spangles Test).
Copper (Cu²⁺)	1. Salt solution + Potassium Ferrocyanide K ₄ [Fe(CN) ₆]. 2. Salt solution + excess NH ₄ OH.	1. Chocolate brown precipitate forms. 2. Pale blue solution turns Deep Royal Blue color [Cu(NH ₃) ₄] ²⁺ .
Aluminum (Al³⁺)	Blue Lake Test: Dissolve white ppt in dil. HCl, add 1-2 drops of Blue Litmus solution, then add	A blue floating precipitate ('Blue Lake') appears suspended in a colorless solution.

	NH ₄ OH dropwise.	
Iron (Fe³⁺)	1. Salt solution + Potassium Ferrocyanide K ₄ [Fe(CN) ₆]. 2. Salt solution + Potassium Thiocyanate (KSCN).	1. Deep Prussian Blue precipitate / color forms. 2. Blood red coloration forms.
Zinc (Zn²⁺)	1. Salt solution + Sodium Hydroxide (NaOH) dropwise to excess. 2. Salt solution + Potassium Ferrocyanide K ₄ [Fe(CN) ₆].	1. White gelatinous ppt forms, soluble in excess NaOH. 2. Bluish-white precipitate forms.
Nickel (Ni²⁺)	Salt solution + NH ₄ OH + Dimethylglyoxime (DMG) reagent.	Brilliant cherry-red / crimson red precipitate forms.
Cobalt (Co²⁺)	Salt solution + Ammonium Thiocyanate (NH ₄ SCN) + acetone/ether.	Beautiful deep blue color in the organic/alcohol layer.
Magnesium (Mg²⁺)	Magneson Reagent Test: Salt solution + NaOH + Magneson reagent.	Blue precipitate / blue absorption complex forms.

--- End of Qualitative Inorganic Salt Analysis Reference Sheet ---